# Sequence and Series Quiz

## Quiz Set 1

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{n}{n^3+4}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \frac{3}{5^n}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{2n+1}{5n+3}\right)^n$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^{5/2}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{1}{2^n+n}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{n^3}{3^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+1}{n^4+n^2+5}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)(n+2)}}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{-n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{n+5}{n^2+2n+3}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 2

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{\sin^2(n)}{n^2+1}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+1}{2^n}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n2^n+1}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{3n+2}{2n+1}\right)^n$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{2n+3}{n^3+n+1}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{n!3^n}{(2n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n+1}{3n+2}\right)^{2n}$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+5}}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{5^n}{n^2 n!}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{n-1}{n^2\sqrt{n}+1}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^{n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{(2n)!}{4^n(n!)^2}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{\cos^2(n)}{n^3+2}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 3

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{n+1}{n^3-n+2}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{n^5}{4^n}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \frac{2^{n+1}}{3^{n-1}}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{1}{3^n+n^2}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \frac{1}{(\ln(n+1))^n}$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^4}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-\sqrt{n}}{n}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{3^n n!}{(n+1)^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{4n-1}{3n+2}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{n^3+2}{n^5+4n^2+1}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{(n!)^3}{(3n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n^2+1}{2n^2+3}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{n2^n}{3^n}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^3}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 4

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{10^n}{n!}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \frac{5}{2^n}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^2+\sqrt{n}}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{n+1}\right)^n$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^{1/3}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{2n^2+3n}{n^4+5n+2}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{n^4}{2^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(1+\frac{2}{n}\right)^{-n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+4n+1}}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{(3n)!}{(n!)^3 3^{3n}}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{\arctan(n)}{n^2+1}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \frac{n^n}{(3n+1)^n}$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{n!}{5^n n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{n+2}{n2^n}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 5

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{n^2}{n^4+3}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{2^n n!}{(2n)!}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=0}^{\infty} \frac{4^{n-1}}{7^n}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(n+1)}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n+2}{4n+1}\right)^n$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{3^n+1}{4^n-2}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+3n}{5^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{5n+1}{2n+3}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{n+1}{n\sqrt{n^2+1}}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{4^n(n!)^2}{(2n+1)!}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+n+1}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(1-\frac{2}{n}\right)^{n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{n^n}{3^n n!}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^2+\ln(n)}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 6

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{n+3}{n^3+2n+1}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{n^{10}}{2^n}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{2n}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+n^2+1}}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n^2+2n}{3n^2+1}\right)^n$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt[4]{n}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+3)}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{(n+1)!}{e^n n!}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{\ln(n)}{n}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{\sin^2(3n)}{n^3+n}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{3^n n^2}{n!}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{1}{5^n+\sqrt{n}}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(\frac{3n+1}{n+4}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{(2n+1)!}{n!n^n}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{n^2-1}{n^4+n^2+2}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 7

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^4+3n+2}}$ is _____ (convergent/divergent).

**Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{n^3+2}{4^n}$ is _____ (convergent/divergent).

**Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \frac{10}{3^n}$ is $S =$ _____.

**Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{n+1}{2n^3+5}$ is _____ (convergent/divergent).

**Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{2n}{3n+1}\right)^{n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is _____ (convergent/divergent).

**Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{\cos^2(2n)}{n^2+4}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{n!n!}{(n+2)!2^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(1+\frac{3}{n}\right)^{-n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{2^n+5}{3^n+1}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{6^n}{n!+1}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{\sqrt{n}+2}{n^2+5n}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n+3}{2n+5}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{(n!)^2}{3^n(2n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 8

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+4}{n^5+n^3+1}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{2^{2n}n!}{(2n)!}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=0}^{\infty} \frac{3^{n+2}}{5^n}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n2^n+3^n}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{6n-1}{2n+1}\right)^n$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^{0.95}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{n}{n^3-\sin^2(n)}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+5}{3^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \frac{1}{(\arctan(n))^n}$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \left(\sqrt{n^2+1} - n\right)$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{(n+2)!}{5^n n!}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{3n+2}{n^4+1}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n^3+1}{2n^3+5}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{n^n}{(2n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n+2}}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 9

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{n+4}{n^3+n^2}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{7^n n^3}{n!}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \frac{7}{6^n}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{3n+4}{5n+1}\right)^n$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^{\pi/2}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{\tan(1/n)}{n}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{(2n)!}{3^n n! n!}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{4n+1}{n+2}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} (1 - \cos(1/n))$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{n+1}{2^n(n^2+1)}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+2n}{n^4+\sqrt{n}}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(1 - \frac{3}{n}\right)^{n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{n^2 4^n}{(n+1)!}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(2n+1)}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 10

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{2^n+n}{4^n-1}$ is _____ (convergent/divergent).

**Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(n^2)!}$ is _____ (convergent/divergent).

**Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=0}^{\infty} \left(\frac{4}{9}\right)^n$ is $S =$ _____.

**Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{n+1}{n^2\sqrt{n+1}}$ is _____ (convergent/divergent).

**Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{2n+5}{7n+2}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^2}}$ is _____ (convergent/divergent).

**Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{n^3+4}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{3^n (n!)^2}{(2n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n+1}{n+3}\right)^{n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^2+3^n}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{n^6}{5^n}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{n+2}{n^3+\cos^2(n)}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(\frac{5n-2}{2n+1}\right)^{2n}$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{n! 2^n}{n^n}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^5+2n}}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 11

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{2n^2+1}{n^4+n+3}$ is _____ (convergent/divergent).

**Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2 5^n}$ is _____ (convergent/divergent).

**Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \frac{2 \cdot 5^n}{8^n}$ is $S =$ _____.

**Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{\sin^2(n)}{n\sqrt{n}}$ is _____ (convergent/divergent).

**Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n}{4n-1}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^{e/2}}$ is _____ (convergent/divergent).

**Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{1}{4^n+n^3}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+2n+1}{3^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{3n+1}{3n+4}\right)^{n^2}$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{n-1}{n^3+2n}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{n!}{(2n)^n}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{\sqrt{n}+1}{n^2+n}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(\frac{2n^2+1}{n^2+5}\right)^{n}$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{8^n(n!)^3}{(3n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$ is _____ (convergent/divergent).

**Justification / Work:**

# Sequence and Series Quiz

## Quiz Set 12

**Name:** _____ **Roll No.:** _____

**Branch:** _____ **Date:** _____

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1. The infinite series $\sum_{n=1}^{\infty} \frac{n^2}{n^5+10}$ is _____ (convergent/divergent).

   **Justification / Work:**

2. The infinite series $\sum_{n=1}^{\infty} \frac{n^3+1}{(n+1)!}$ is _____ (convergent/divergent).

   **Justification / Work:**

3. The sum of the infinite geometric series $\sum_{n=1}^{\infty} \frac{1}{4^{n-1}}$ is $S =$ _____.

   **Justification / Work:**

4. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+100}}$ is _____ (convergent/divergent).

   **Justification / Work:**

5. The infinite series $\sum_{n=1}^{\infty} \left(\frac{\sqrt{n}}{\sqrt{n}+1}\right)^{n\sqrt{n}}$ is _____ (convergent/divergent).

   **Justification / Work:**

6. The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^{7/3}}$ is _____ (convergent/divergent).

   **Justification / Work:**

7. The infinite series $\sum_{n=1}^{\infty} \frac{e^{-n}}{n^2+1}$ is _____ (convergent/divergent).

**Justification / Work:**

8. The infinite series $\sum_{n=1}^{\infty} \frac{(2n+2)!}{(n!)^2 4^n}$ is _____ (convergent/divergent).

**Justification / Work:**

9. The infinite series $\sum_{n=1}^{\infty} \left(\frac{n+4}{5n+2}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

10. The infinite series $\sum_{n=1}^{\infty} \frac{n+10}{n^3+5n}$ is _____ (convergent/divergent).

**Justification / Work:**

11. The infinite series $\sum_{n=1}^{\infty} \frac{n^2+4n}{2^n n!}$ is _____ (convergent/divergent).

**Justification / Work:**

12. The infinite series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n^2+4)}}$ is _____ (convergent/divergent).

**Justification / Work:**

13. The infinite series $\sum_{n=1}^{\infty} \left(\frac{7n+1}{3n+2}\right)^n$ is _____ (convergent/divergent).

**Justification / Work:**

14. The infinite series $\sum_{n=1}^{\infty} \frac{5^n (n!)^2}{(2n)!}$ is _____ (convergent/divergent).

**Justification / Work:**

15. The infinite series $\sum_{n=1}^{\infty} \frac{2^n+3^n}{5^n-1}$ is _____ (convergent/divergent).

**Justification / Work:**